

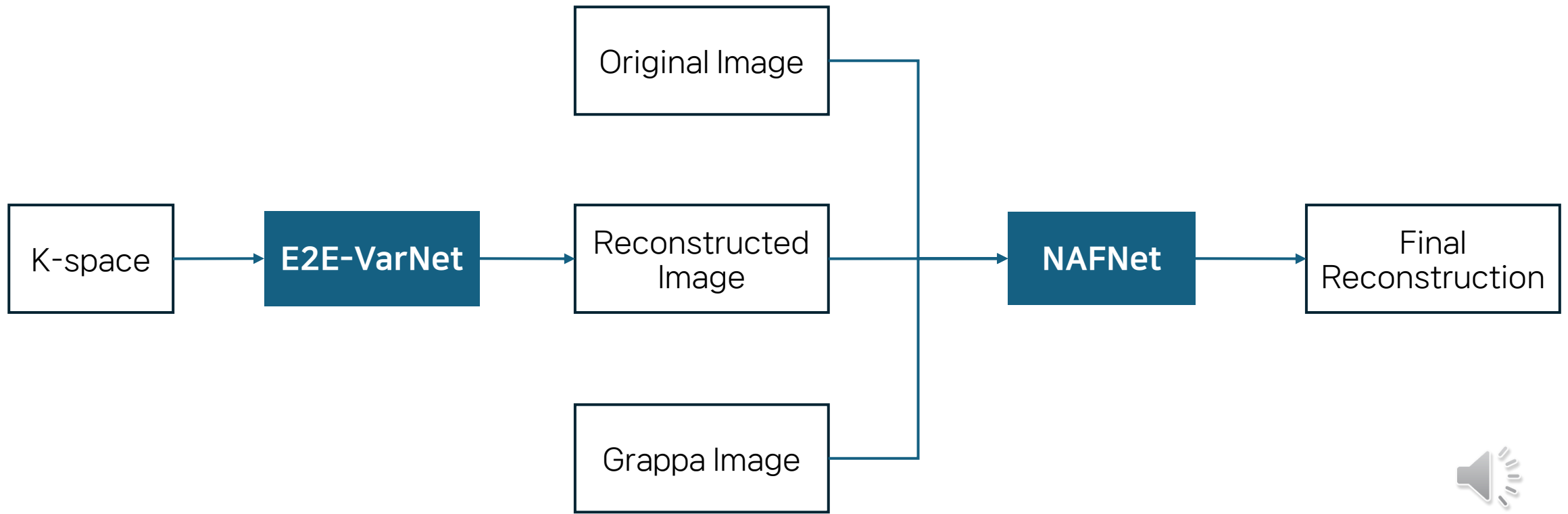
2024 SNU FastMRI Challenge

Team 시시시 - 수상소감

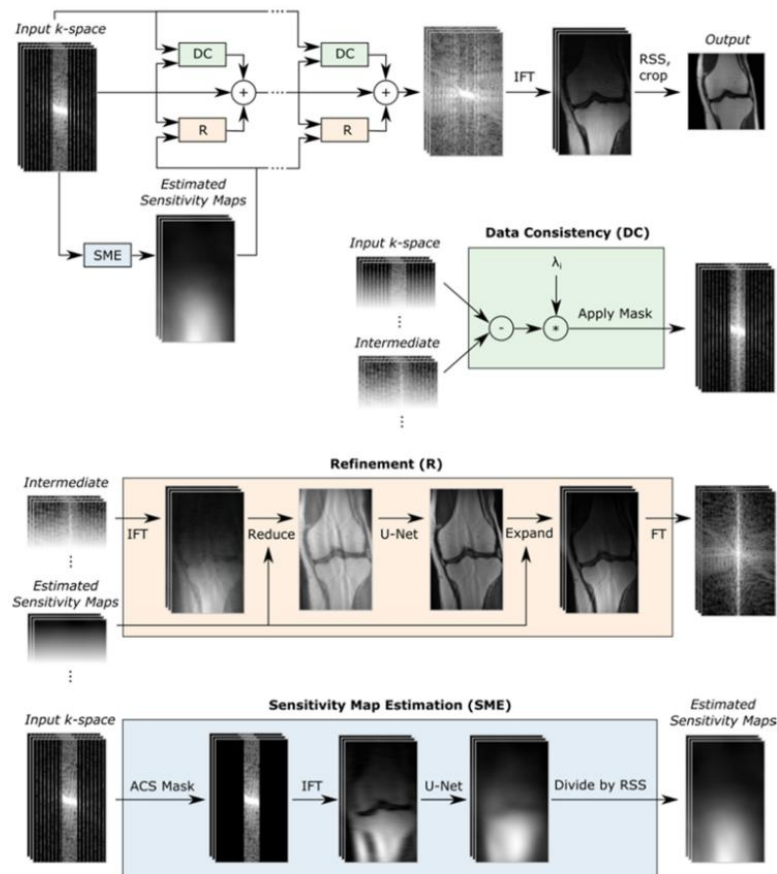
김진수, Computer Science and Engineering
박정민, Computer Science and Engineering



Overall Model Architecture



E2E-VarNet



Block diagram of E2E-VarNet [1]

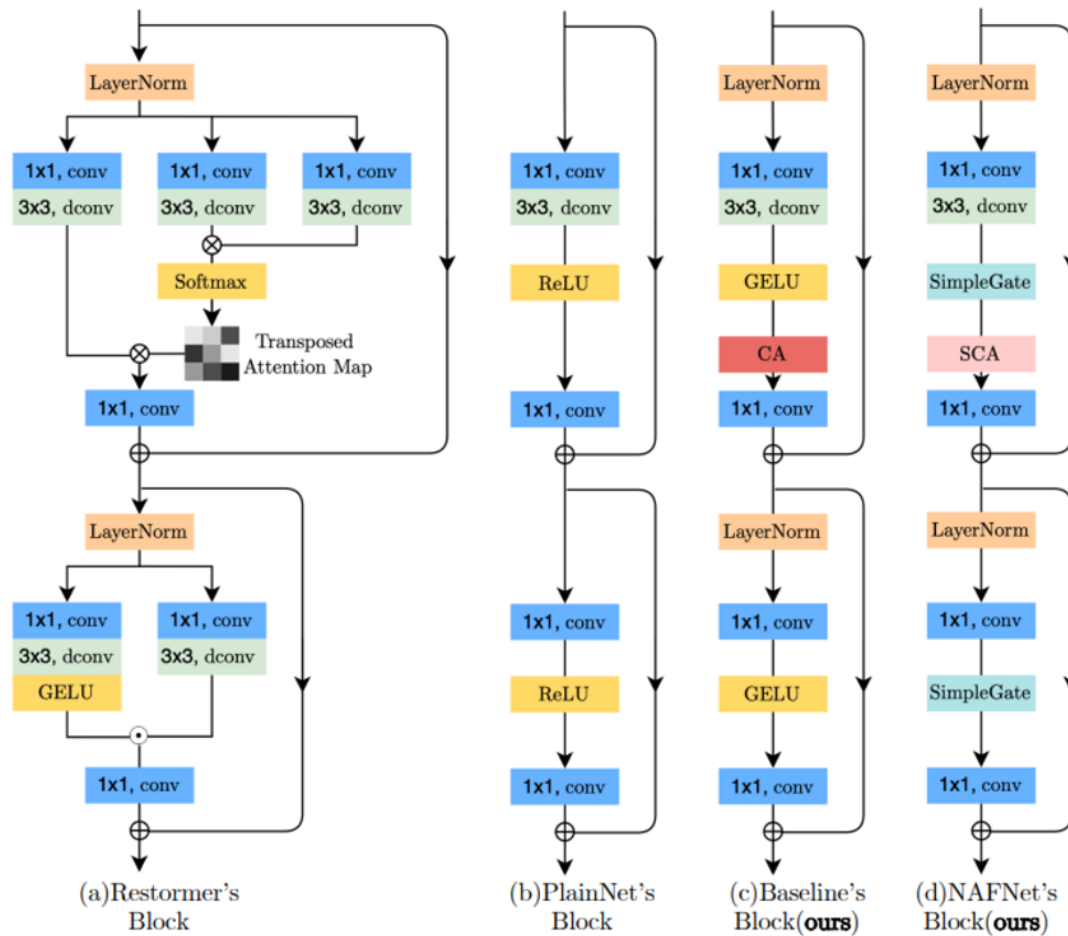
- K-space to Image (**End-to-End**) Model
- Consists of cascade layers

$$\mathbf{k}^{t+1} = \mathbf{k}^t \left[\underbrace{-\eta^t M(\mathbf{k}^t - \tilde{\mathbf{k}})}_{\text{Data Consistency}} \right] \underbrace{+ G(\mathbf{k}^t)}_{\text{Refinement}}$$

$$G(\mathbf{k}^t) = \mathcal{F} \circ \underbrace{\mathcal{E}}_{\text{Provided by SME}} \circ \text{CNN}(\mathcal{R} \circ \mathcal{F}^{-1}(\mathbf{k}^t)).$$

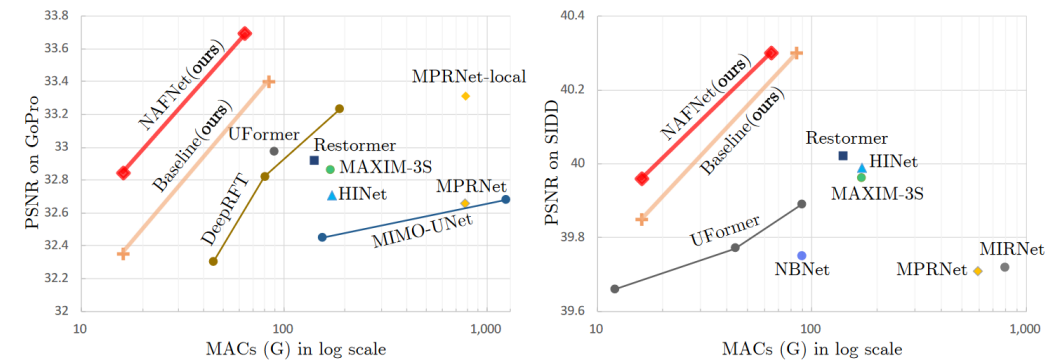
Hparams for Varnet structure
: (cascade, chans, sens_chans) = (6, 15, 4)

NAFNet



Intra-block structure of NAFNet and other models [2]

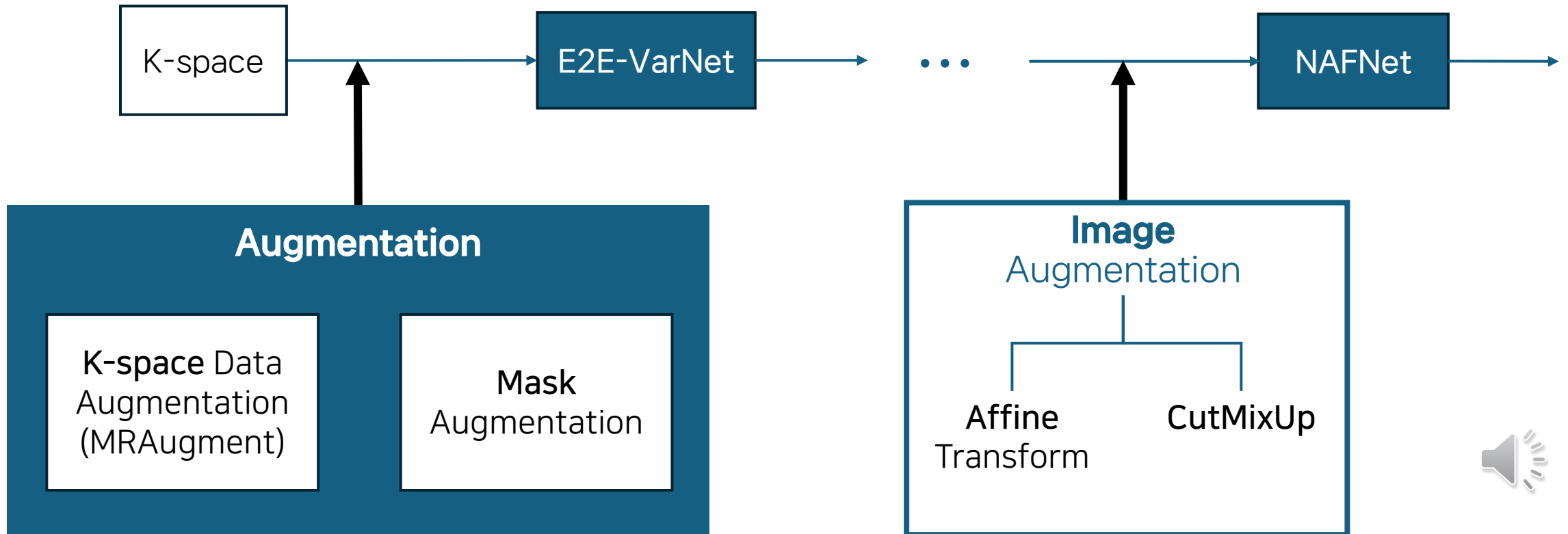
- Image Denoising/Deblurring model
- **Lower Complexity** than other models



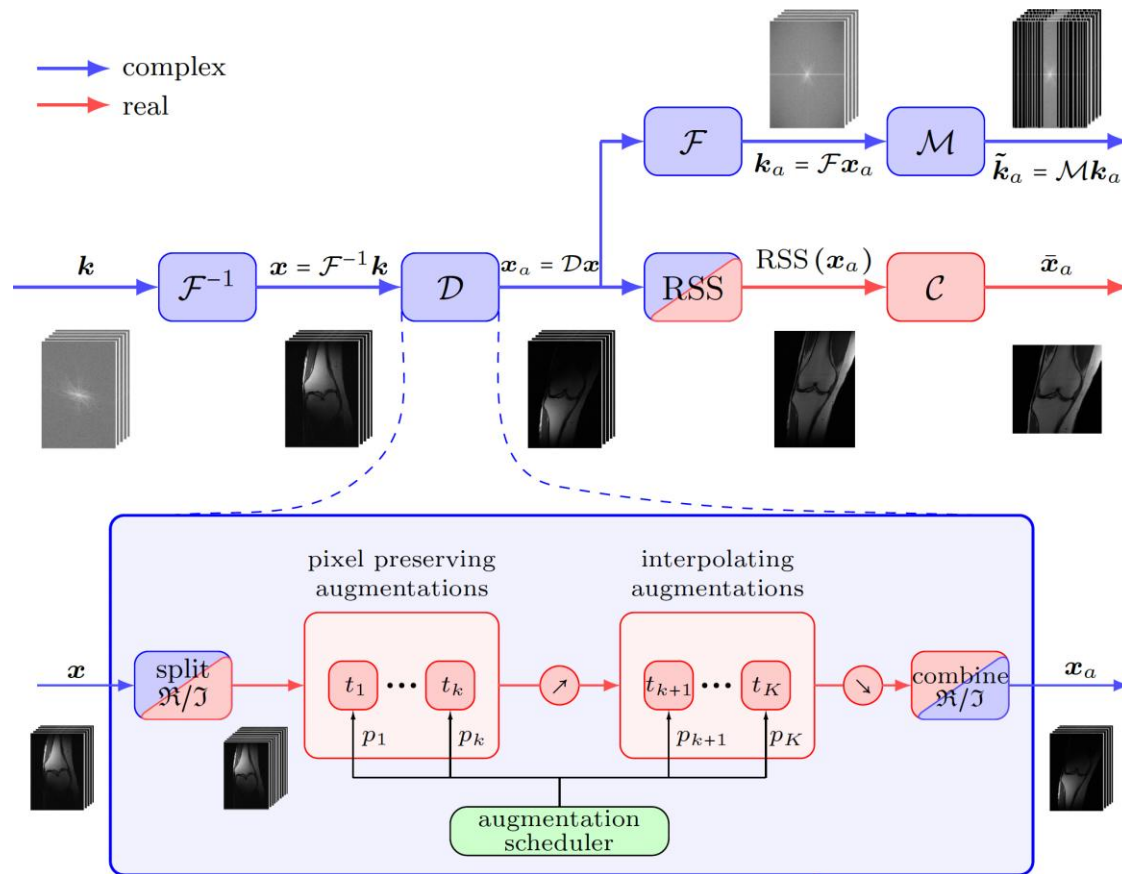
- 3-channel Input
 - Original Image
 - Reconstructed Image from Varnet
 - Grappa Image



Data Augmentation



K-space: MRAugment



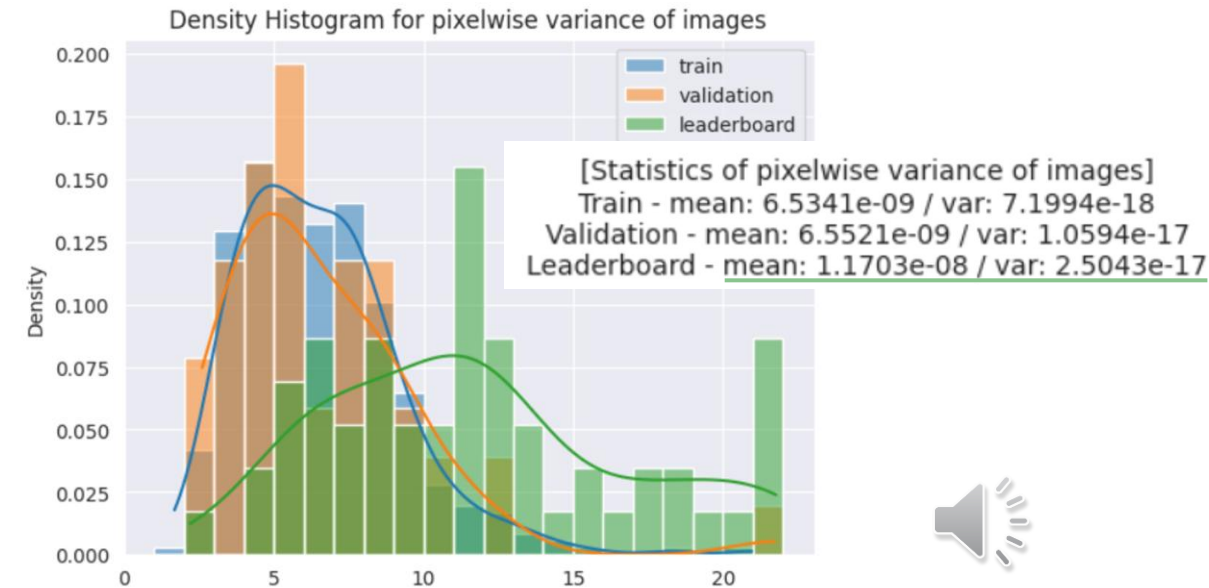
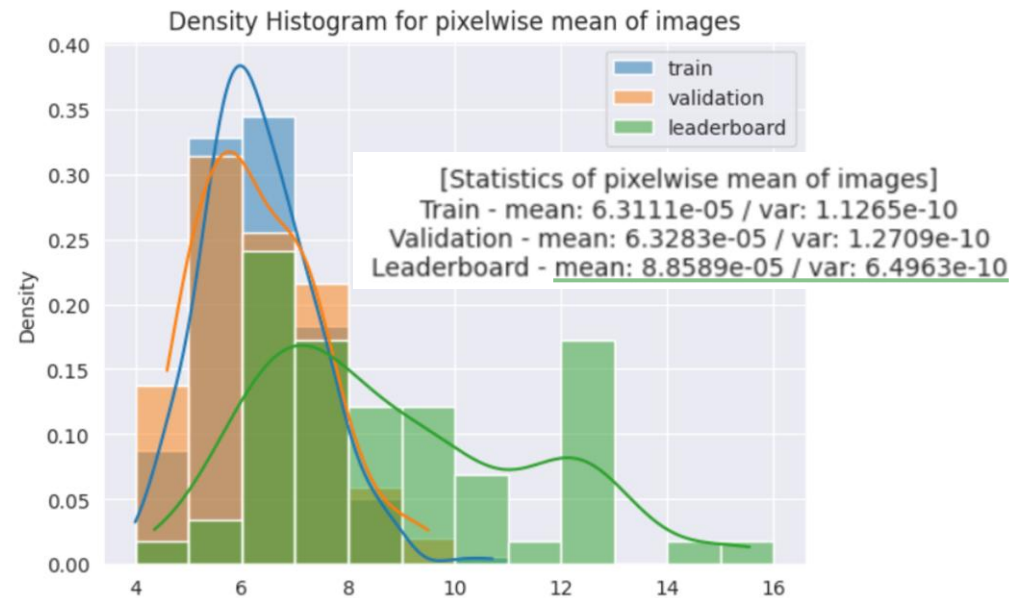
- Pixel preserving Augmentations
 - : Horizontal/Vertical flip, Translation (by integer number of pixel)
- Interpolating Augmentations
 - : Rotation, Shearing, Scaling



K-space: MRAugment

- Color Augmentation (Brightness, Contrast)

: To cover different distribution of pixel values between train/val data and LB data

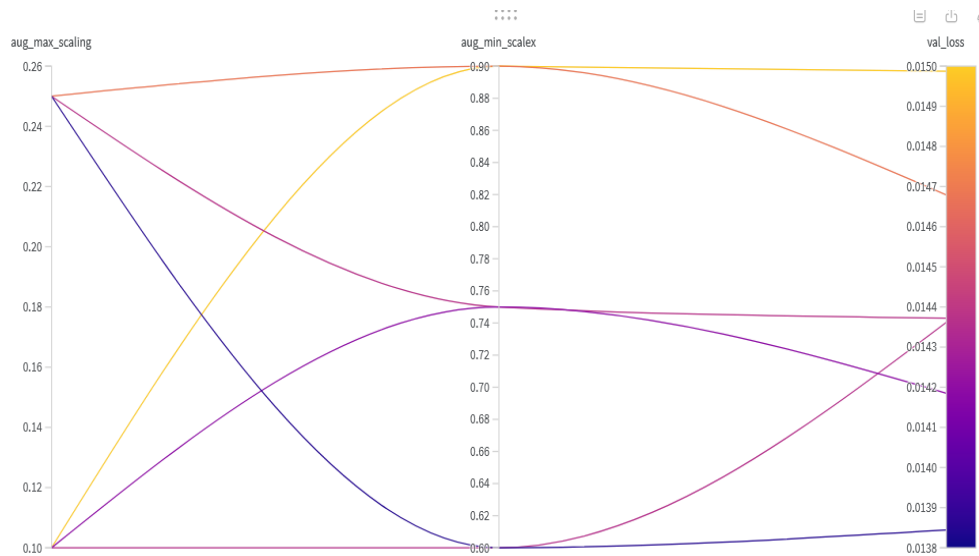


Density Histogram for pixelwise mean/variance of target images in datasets

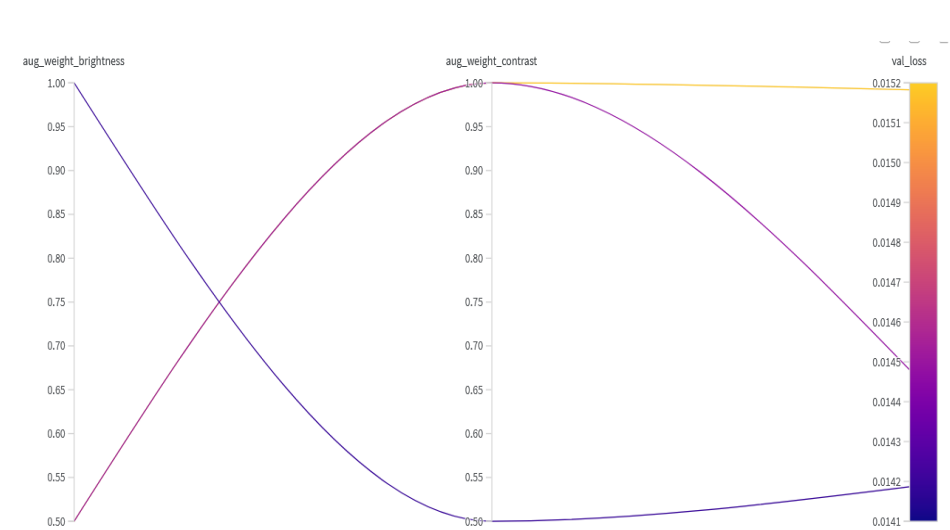


K-space: MRAugment

- **Random Search with WandB Sweep:** To find optimal combination of hyperparameters



Hparams for Scaling
(aug_max_scaling, aug_min_scalex)



Hparams for color augmentation
(aug_weight_brightness, aug_weight_contrast)



Mask Augmentation

```
mask_augmentor = MaskAugmentor(args, lambda: epoch,  
                                center_fractions=[0.08, 0.083],  
                                accelerations=[6, 7, 9],  
                                allow_any_combination=True)
```

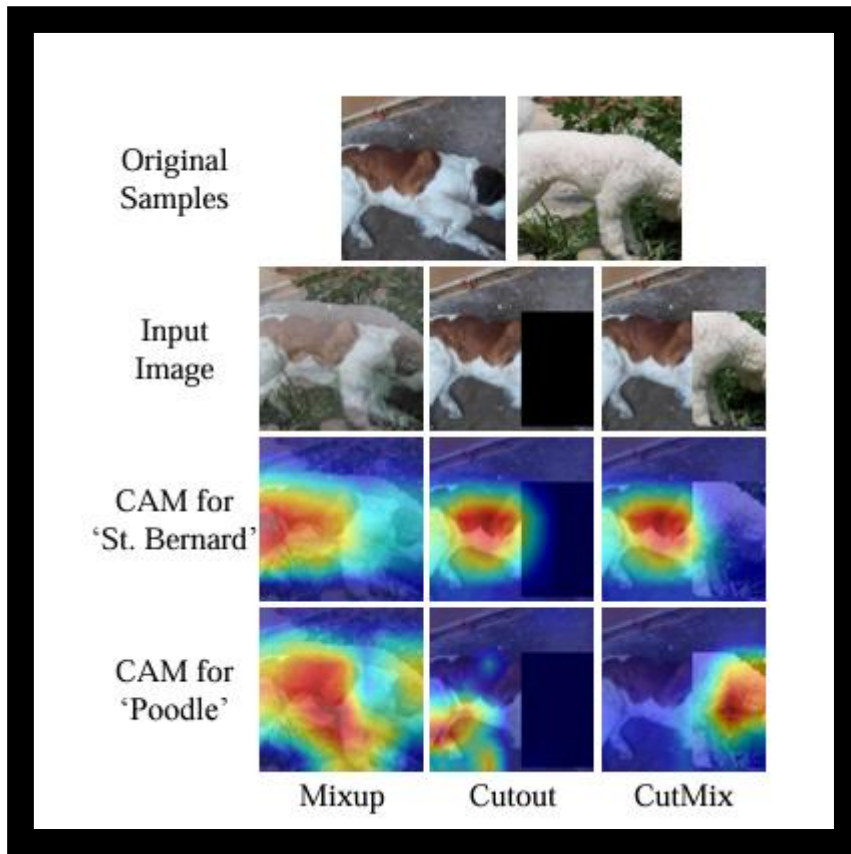
```
if self.low_to_high_acc or p > 0.0 and self.random_apply():  
    mask, _ = self.mask_func(shape)  
    mask = mask.byte()  
else:  
    mask = original_mask
```

```
self.mask_func = EquiSpacedMaskFunc(center_fractions, accelerations, allow_any_combination)
```

- **Private acc(x5)** data is given from 2024 challenge (cf. ONLY public acc data in ~2023 challenge,) which require model more **robust** to various accelerations.
- Mask Augmentor creates **private acc(x6, x7, x9) data** from fully-sampled train/val data, which are used to training and expose model to various accelerations.



Image Augmentation



- Affine Transformation + CutMixUp
- **CutMixUp(Cutmix & Mixup)**
 - **CutMix:** Attach cropped image to other image
 - **MixUp:** Mix two different images
 - CutMixUp improved performance on Super Resolution task [5]



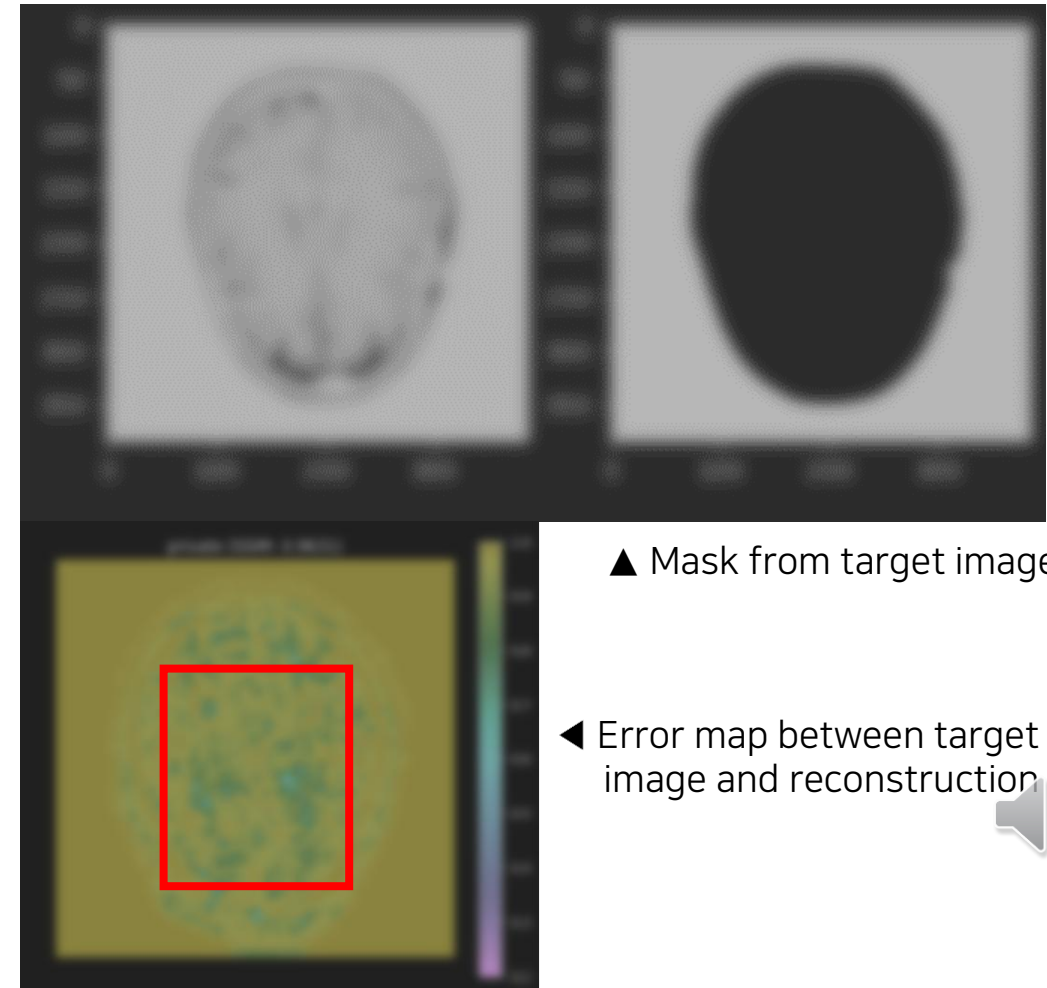
CAM visualizations with Mixup/CutMix [4]

Loss

Loss with Mask

In MRI images, a mask is applied to the brain area, and the loss is calculated only within the masked region. This approach encourages focused learning on the desired region.

As we observed higher error within the brain, the size of the mask was reduced in the later stages of training to further concentrate on the internal areas.



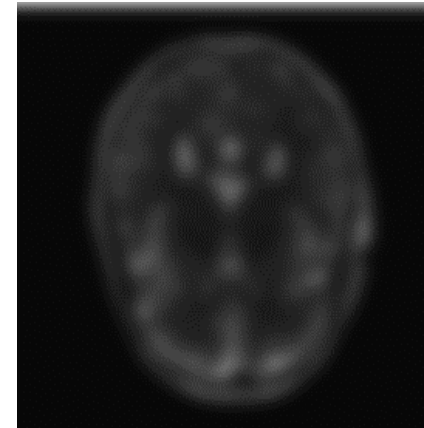
Loss

Index-Based Loss

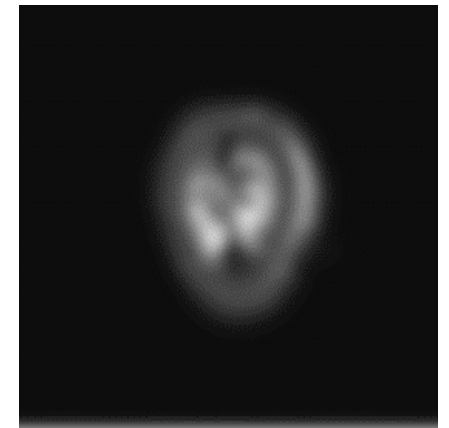
As the slice index increases in MRI data, the size of the mask tends to decrease, and the shapes become relatively simpler, making it easier to achieve high SSIM values.

We weighted the loss calculations based on the area proportional to the mask size, which is assumed to follow $y = \cos^2 x$ ($0 \leq x \leq \frac{\pi}{2}$) graph.

Slice 0



Slice 13



```
def set_weight(self, num_slices):  
    for i in range(num_slices):  
        # assume that the weight is prop to size of mask which is prop to cos^2  
        self.weight_list.append(math.cos(i * math.pi / (num_slices * 2)) ** 2)
```

$$\text{Index-based weighted loss} = \text{SSIM} \times \cos^2\left(\frac{\text{current slice index}}{\text{max slice index}} \times \frac{\pi}{2}\right)$$



Training Steps

- Step 1: Train **E2E-VarNet** 50 epochs
 - lr: 1e-4, loss: index-based weighted SSIM, Optimizer: Adam
 - MRAugment & Mask Augmentation applied
 - No validation
- Step 2: Train **E2E-VarNet** (from step 1) 15 epochs more
 - lr: 1e-4, loss: index-based weighted SSIM, Optimizer: AdamW
 - MRAugment & Mask Augmentation applied



Training Steps

- Step 3: Train NAFNet 10 epochs
 - lr: 1e-4, loss: index-based weighted SSIM, Optimizer: AdamW
 - lr scheduler: CosineAnnealing
 - Image Augmentation applied

- **Exponential Scheduler** is used for scheduling all augmentation

- p_{max} was 0.4(step 1), 0.6(step 2), and 0.5(step 3.)

$$p(t) = \frac{p_{max}}{1 - e^{-c}} (1 - e^{-tc/T})$$



Advancing Process

Description	LB SSIM Score
Baseline (38 epochs)	0.9572
Use Large Varnet	0.9681 (+0.0109)
+ MRAugment, Mask Augmentation	0.9772 (+0.0121)
+ loss only for mask	0.9787 (+0.0015)
+ adjusted MRAugment hparams	0.9788 (+0.0001)
+ index-based loss, no validation	0.9792 (+0.0004)
+ train 15 epochs more	0.9796 (+0.0004)
+ add NAFNet	0.9803 (+0.0007)



아쉬운 점

- Lack of DL training experience
- Insufficient control of variables in random search
- Remaining Experiments
 - different model(KBNet, DIRCN), different loss, ...
- Lack of understanding for K-space data



배운 점

김진수

- 생소한 Domain에 대한 경험
- WandB 등 MLOps 활용 경험
- 데이터 분석의 중요성(train/val, LB)
- 1달 반의 대회 기간 -> 텐션 유지의 중요성

박정민

- Grappa image를 통해 학습결과 역시 비슷한 양상이 될 것을 예측했으나 전혀 다른 결과
- 정교한 실험 설계의 중요성
- 실험 전에 결과를 단정짓지 않고, 모든 가능성을 열어두기
- 장비의 중요성, GPU의 100% 활용



다음 대회에서는...

More general dataset – including knee image

- So far, ONLY brain images were treated in SNU FastMRI Challenge
- Adding knee images to dataset would require additional analysis of dataset and promote participants to make more robust model

How to promote leaderboard update? (with the highest score)

- Even though LB score submission was forced once in a week, the penalty seemed not to be effective (since local GPU can be used)
- Furthermore, lowering score by re-submitting is (implicitly) allowed



감사합니 다

